US 20250278013A1

(19) United States
(12) Patent Application Publication
SATO
(10) Pub. No.: US 2025/0278013 A1
(43) Pub. Date: Sep. 4, 2025

(54) **IMAGE PICKUP APPARATUS AND CAMERA SYSTEM HAVING THE SAME**

(71) Applicant: **CANON KABUSHIKI KAISHA**, Tokyo (JP)

(72) Inventor: **HIDEKI SATO**, Tokyo (JP)

(21) Appl. No.: **19/059,768**

(22) Filed: **Feb. 21, 2025**

(30) **Foreign Application Priority Data**

Mar. 4, 2024 (JP) ................................ 2024-031873

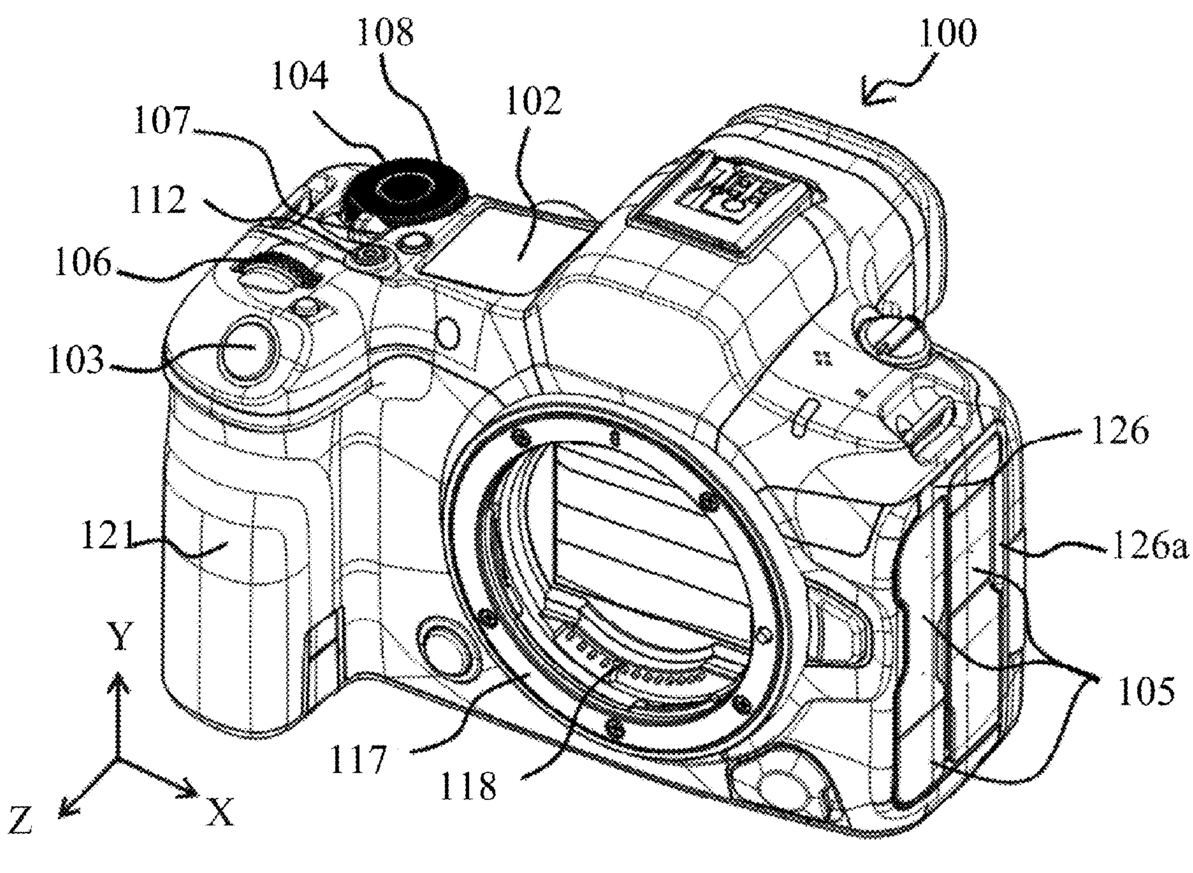
**Publication Classification**

(51) **Int. Cl.**
***G03B 17/55*** (2021.01)
***G03B 17/56*** (2021.01)
***H04N 23/51*** (2023.01)
***H04N 23/52*** (2023.01)

(52) **U.S. Cl.**
CPC .......... ***G03B 17/55*** (2013.01); ***G03B 17/566*** (2013.01); ***H04N 23/51*** (2023.01); ***H04N 23/52*** (2023.01); *G03B 17/563* (2013.01)

(57) **ABSTRACT**

An image pickup apparatus to which a cooling apparatus including a fan is detachable includes a heat generating member, a first exterior member that forms a first surface of a housing and has a first opening, a second exterior member that forms a second surface of the housing and has a second opening, a heat radiating member that is arranged inside the housing to face a surface opposite to the first surface and is configured to cool heat generated by the heat generating member, a first sealant arranged between the first exterior member and the heat radiating member, and a second sealant arranged between the first exterior member and the second exterior member, and between the heat radiating member and the second exterior member. The first and second exterior members and the heat radiating member form a first flow path of air between the first and second openings.



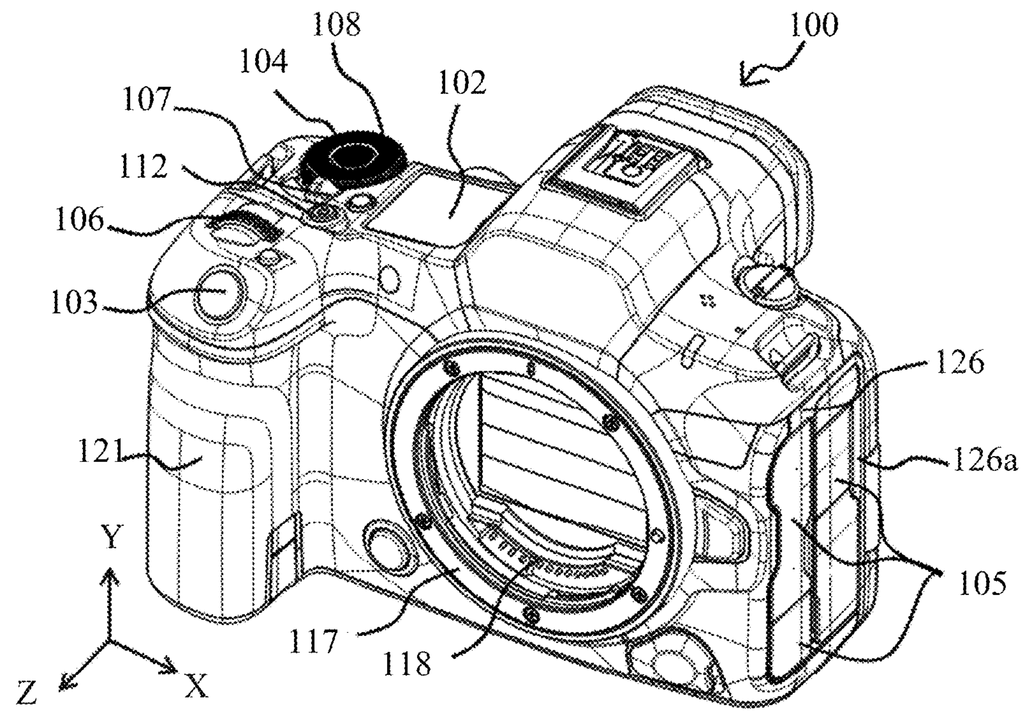

100
108
104
102
107
112
106
103
126
121
126a
Y
105
117
118
Z
X

FIG. 1A

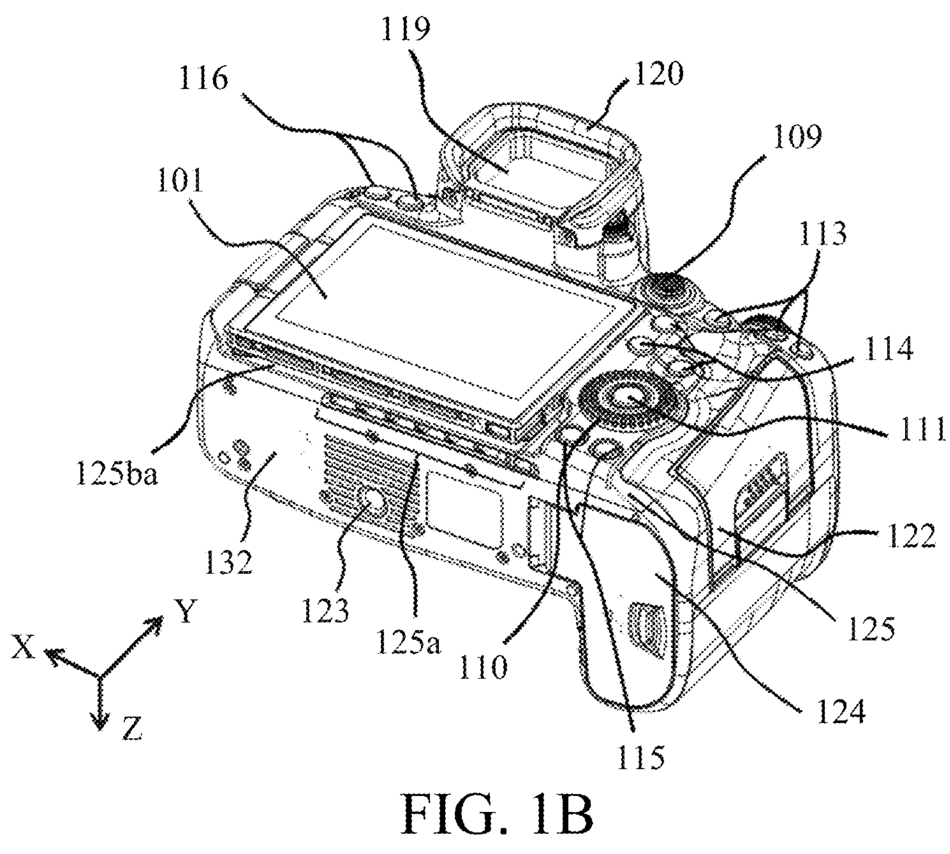

119
116
120
101
109
113
114
111
125ba
122
132
123
125a
125
Y
X
110
124
Z
115

FIG. 1B

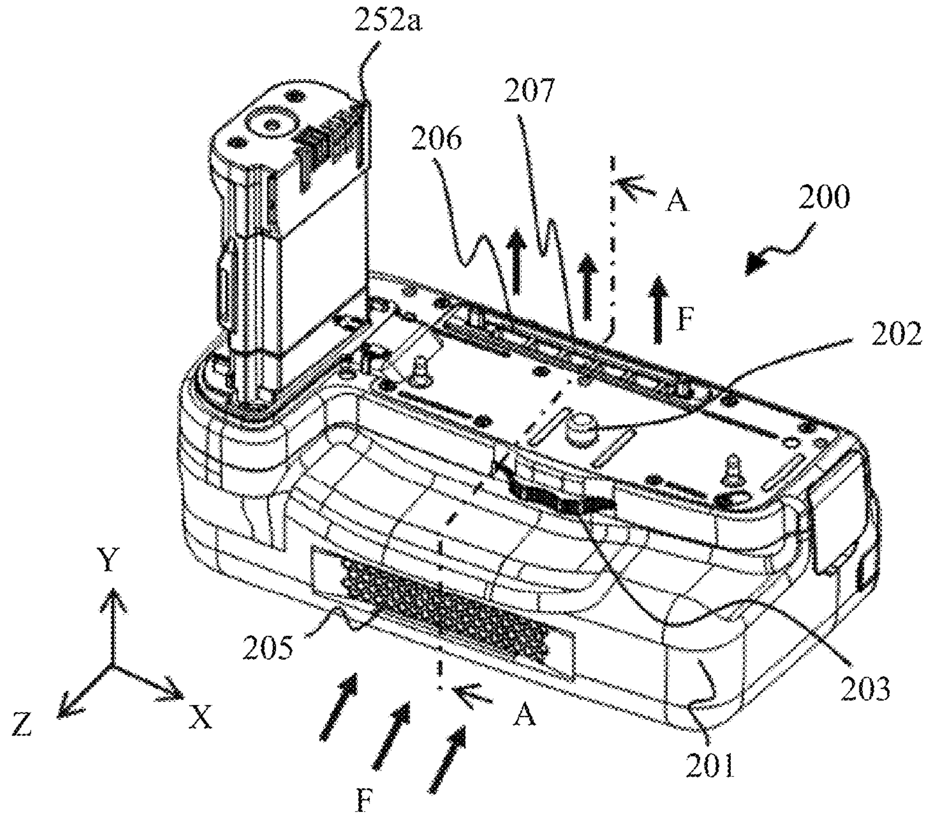

252a
207
206
A
200
F
202
Y
205
203
Z
X
A
F
201

FIG. 2A

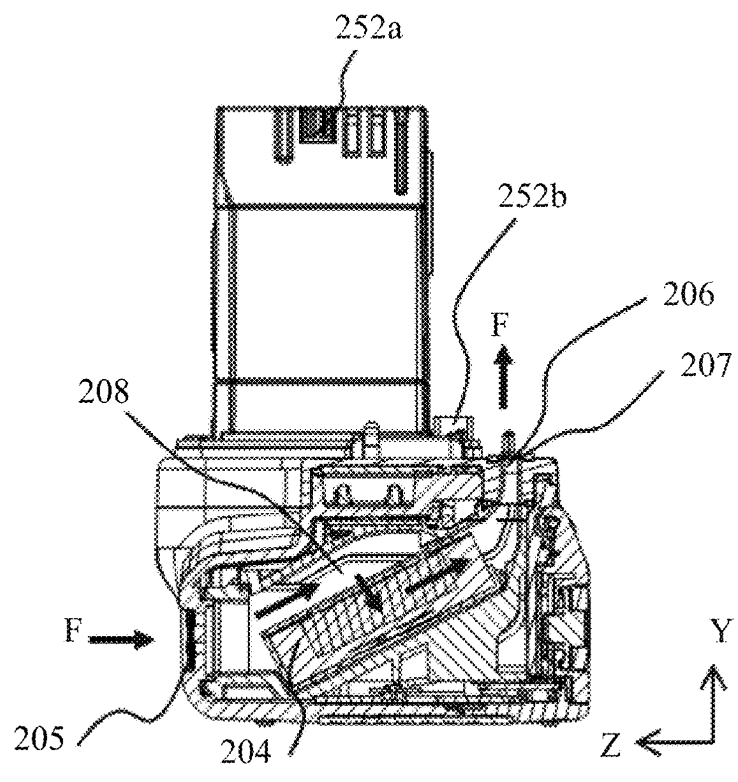

252a
252b
206
F
207
208
F
Y
205
204
Z

FIG. 2B

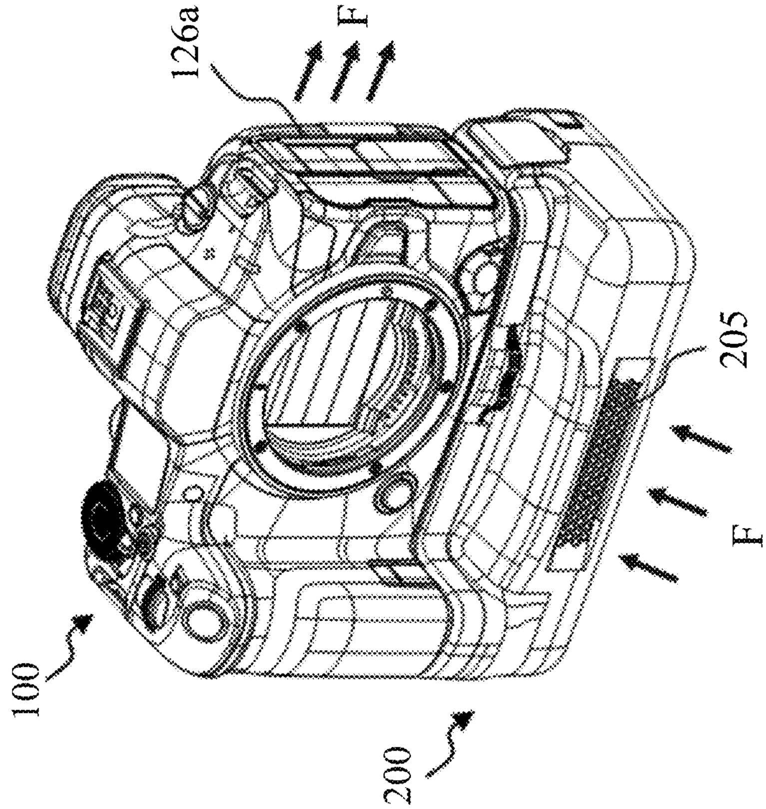



FIG. 3B

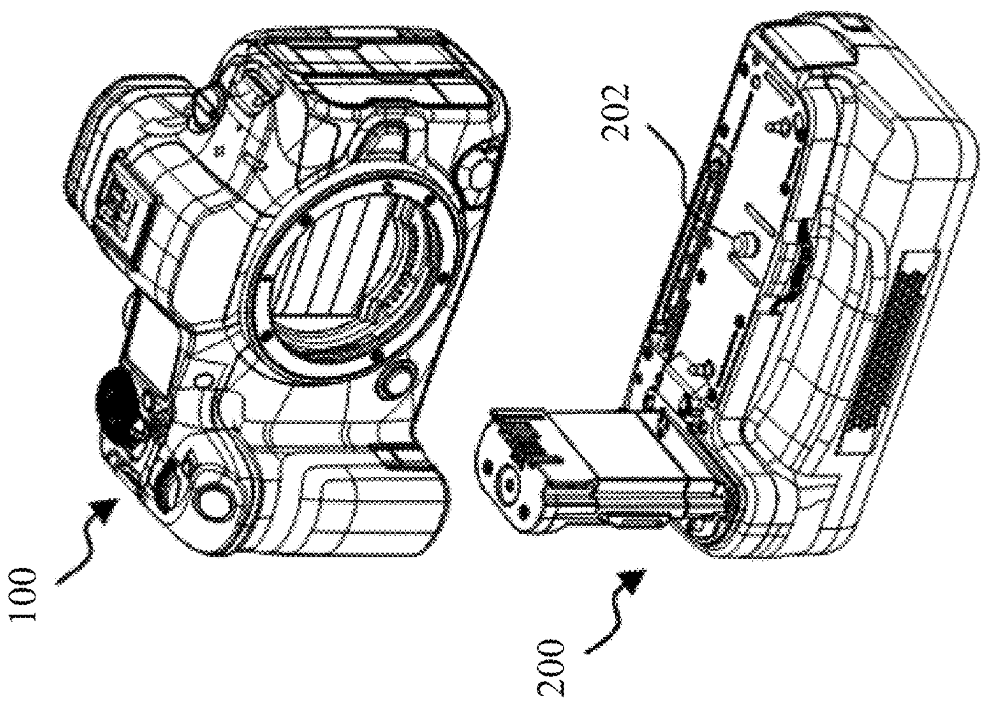



FIG. 3A

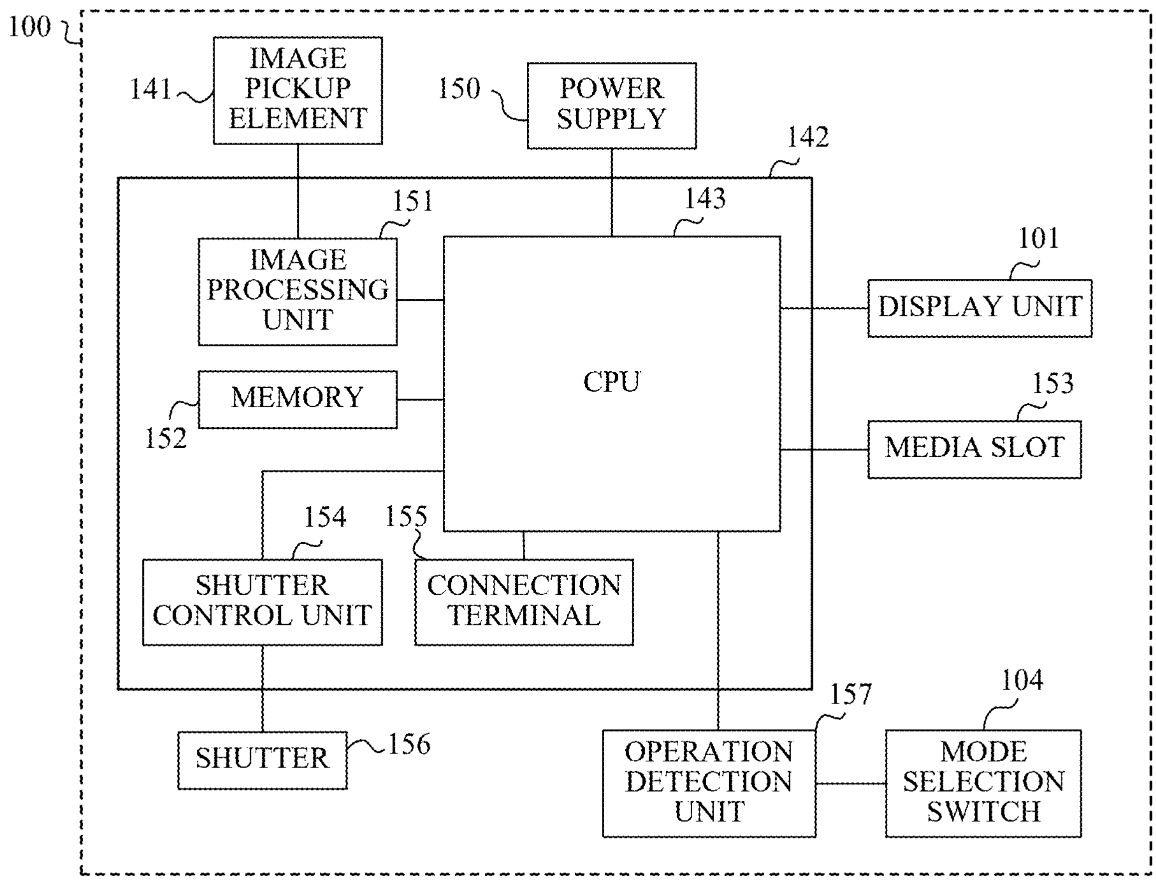

100
141
IMAGE PICKUP ELEMENT
150
POWER SUPPLY
142
151
143
IMAGE PROCESSING UNIT
CPU
101
DISPLAY UNIT
MEMORY
152
153
MEDIA SLOT
154
155
SHUTTER CONTROL UNIT
CONNECTION TERMINAL
157
104
SHUTTER
156
OPERATION DETECTION UNIT
MODE SELECTION SWITCH

FIG. 4A

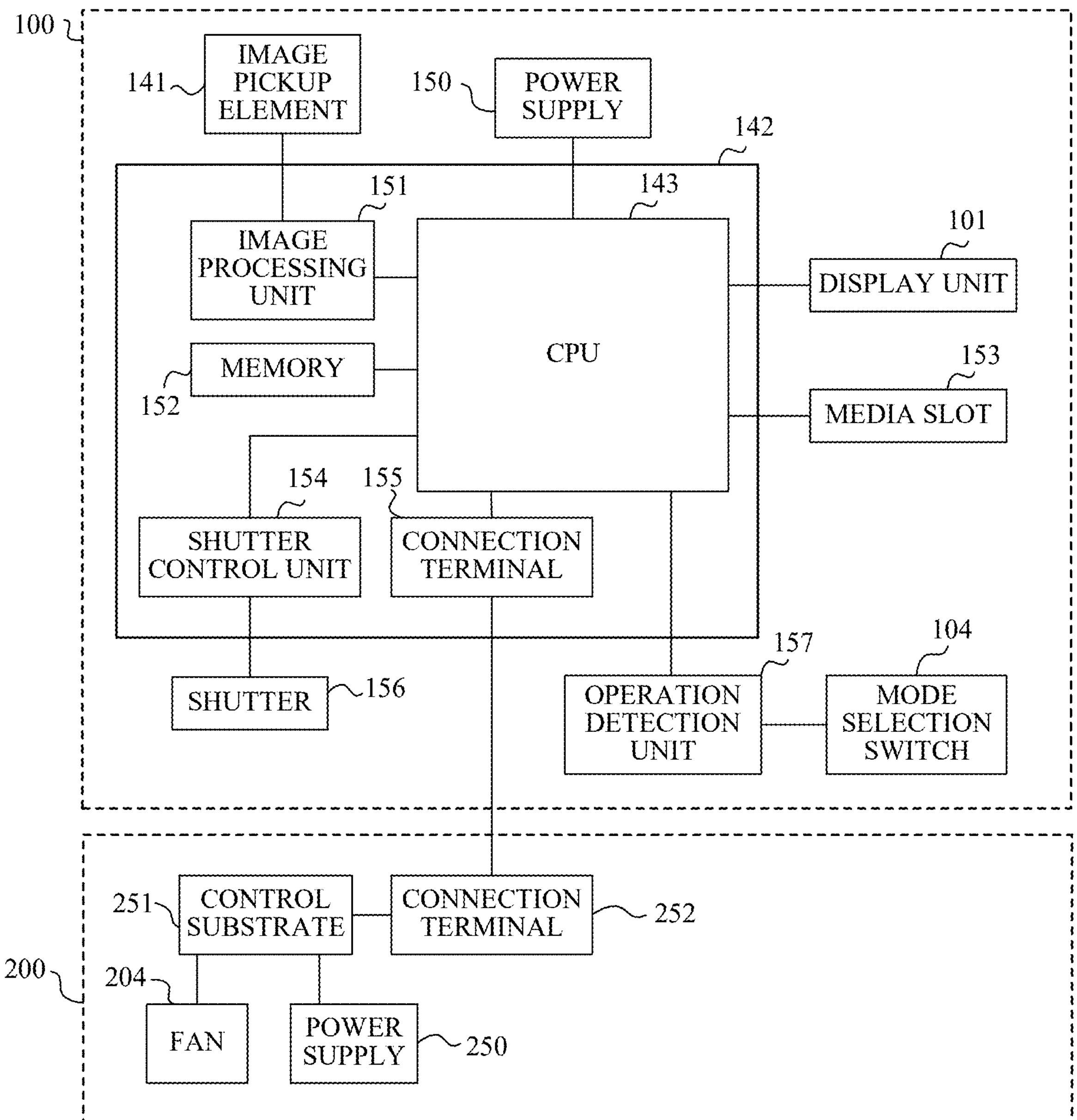

100
141
IMAGE PICKUP ELEMENT
150
POWER SUPPLY
142
151
143
IMAGE PROCESSING UNIT
CPU
101
DISPLAY UNIT
MEMORY
152
153
MEDIA SLOT
154
155
SHUTTER CONTROL UNIT
CONNECTION TERMINAL
157
104
SHUTTER
156
OPERATION DETECTION UNIT
MODE SELECTION SWITCH
251
CONTROL SUBSTRATE
CONNECTION TERMINAL
252
200
204
FAN
POWER SUPPLY
250

FIG. 4B

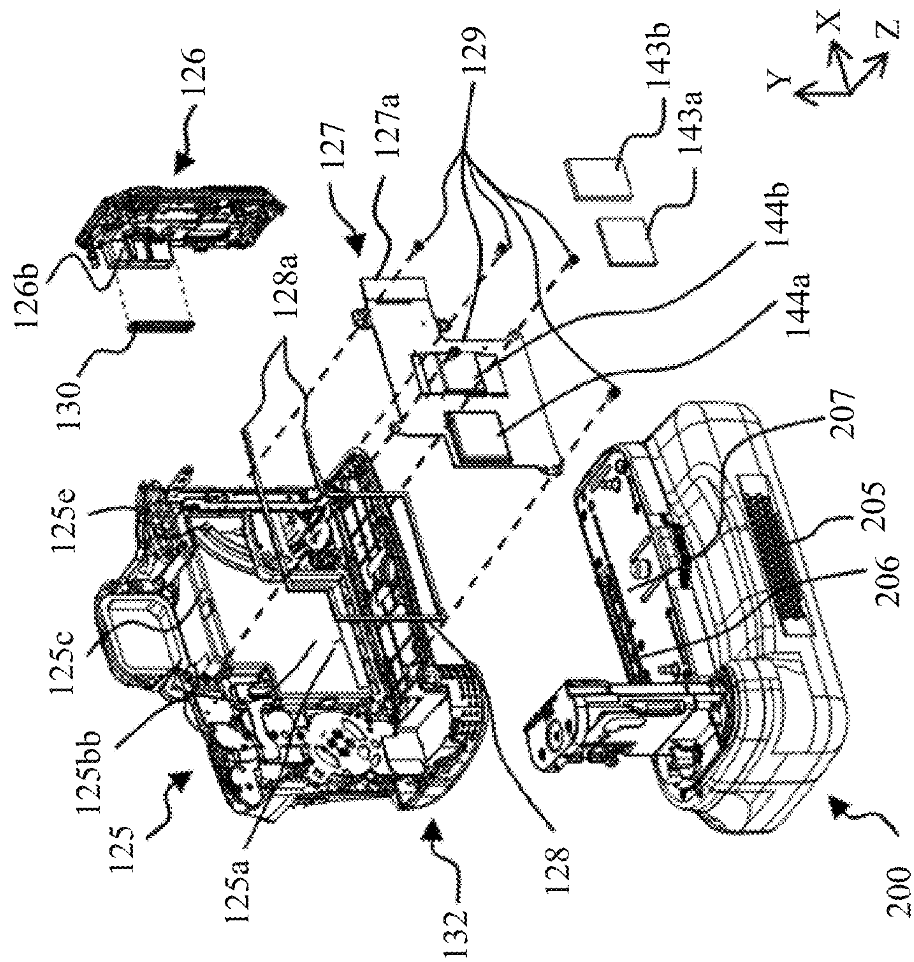

126b
126
130
127
127a
129
128a
125e
125c
125bb
125
125a
132
128
144b
144a
143b
143a
207
205
206
200
Y
X
Z

FIG. 5A

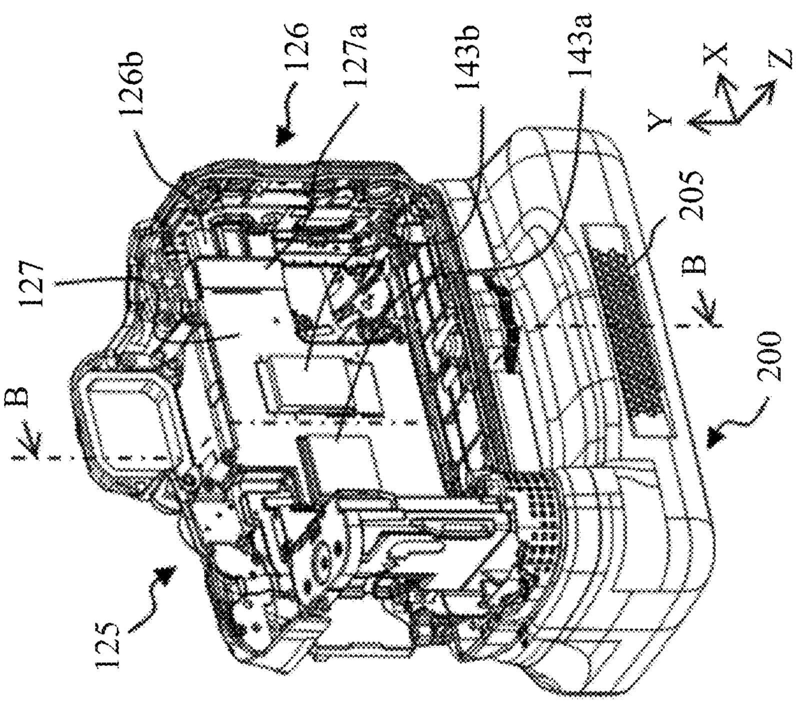

126b
126
127a
143b
143a
127
B
B
205
200
125
Y
X
Z

FIG. 5B

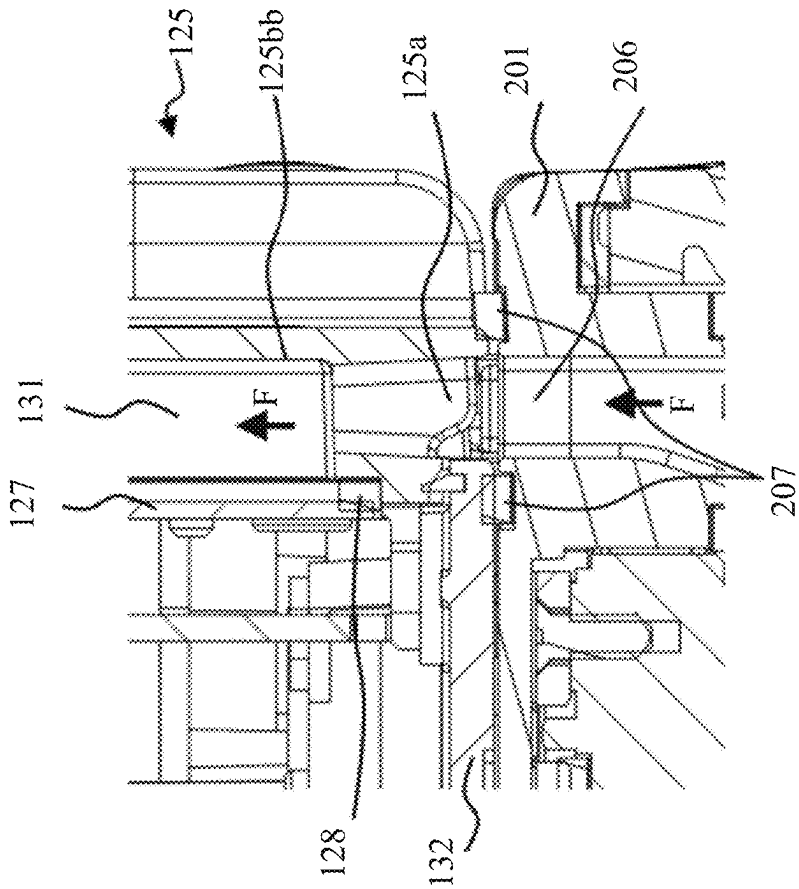

125
125bb
125a
201
206
131
F
F
127
207
128
132

FIG. 5D

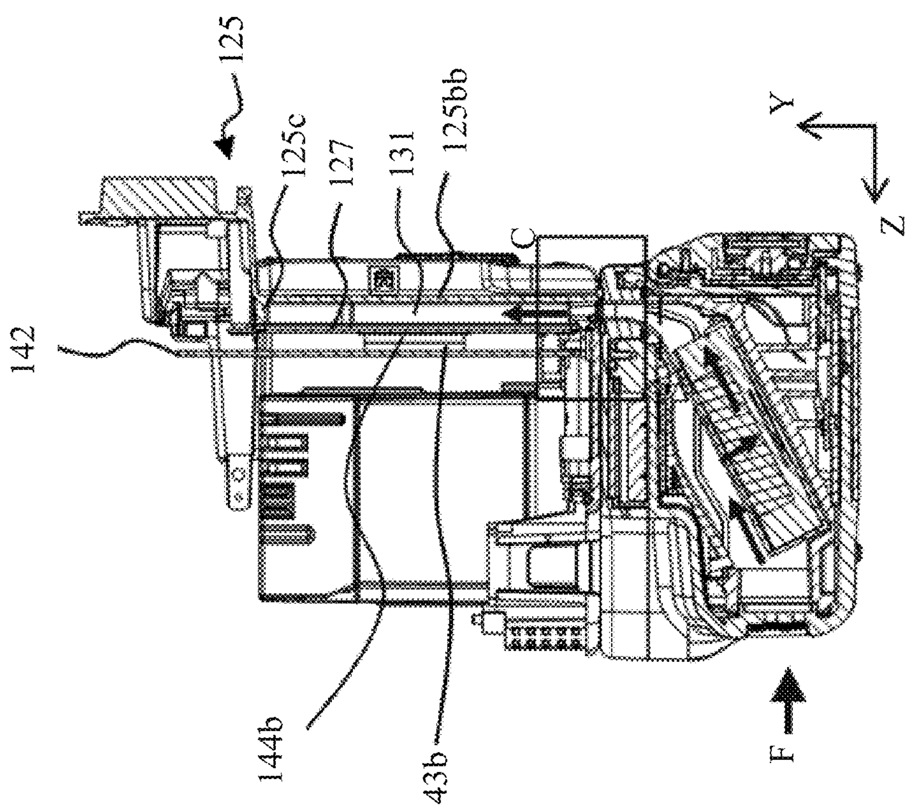

125
125c
127
131
125bb
142
C
Y
Z
144b
143b
F

FIG. 5C

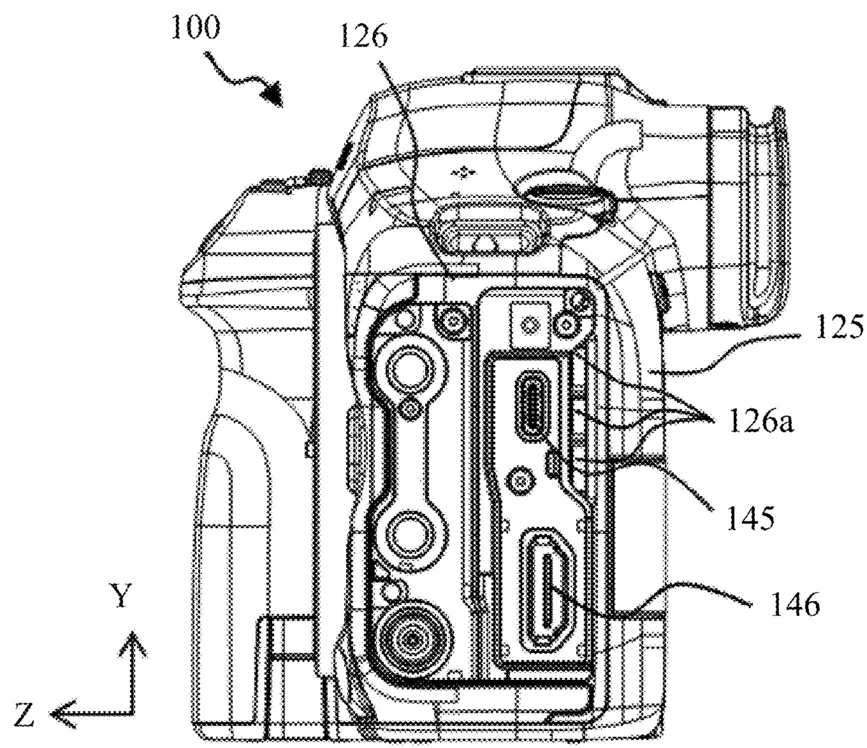



FIG. 6A

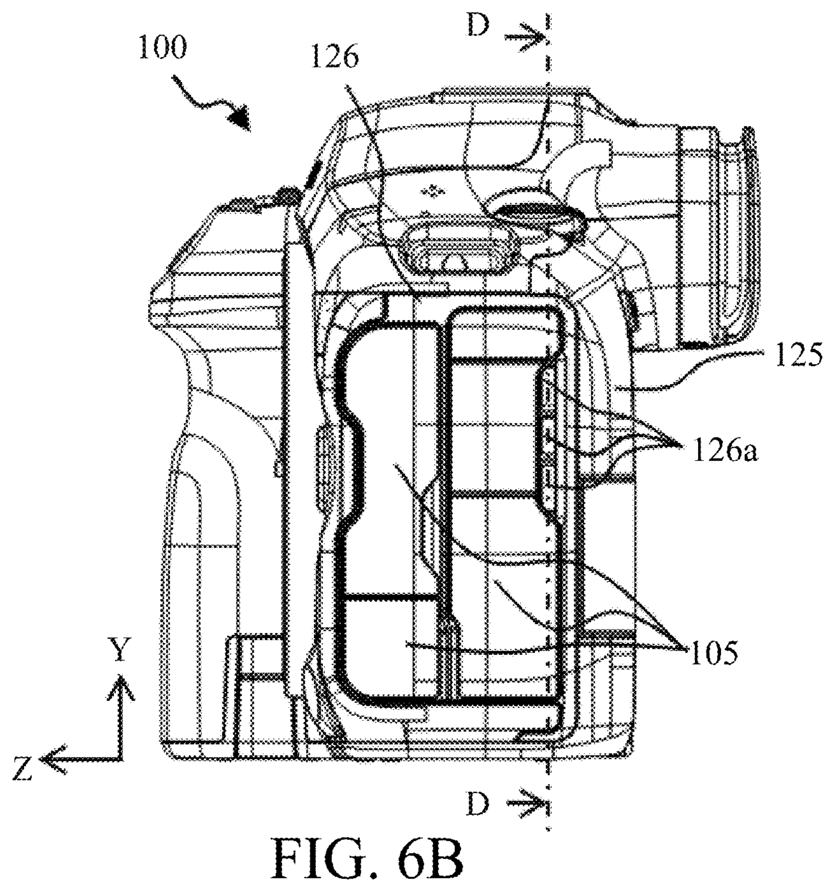



FIG. 6B

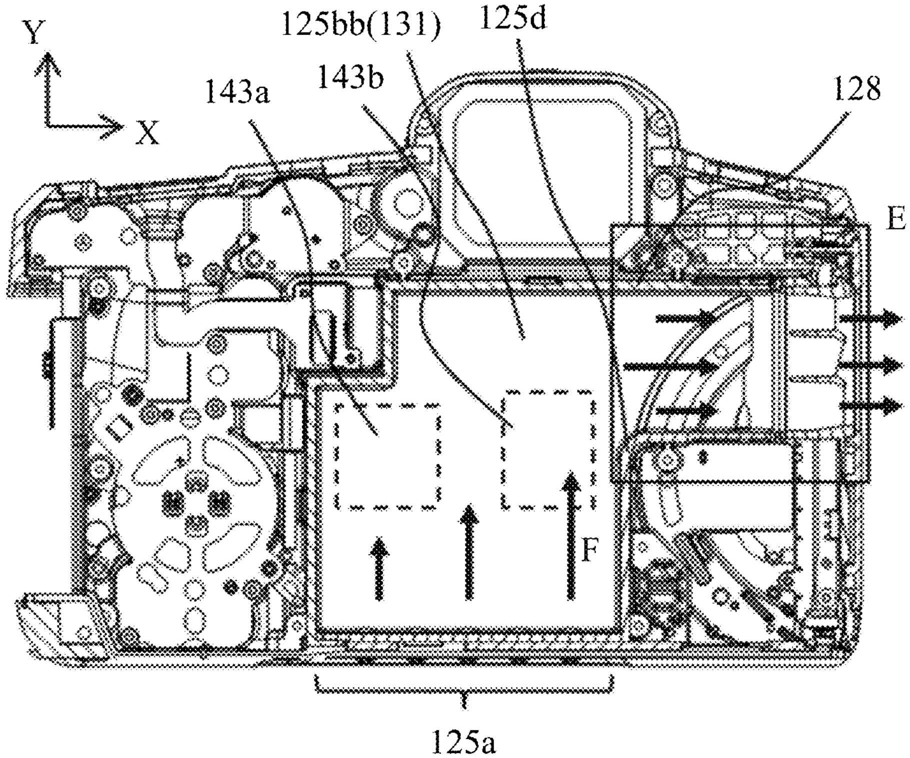

Y
X
143a
143b
125bb(131)
125d
128
E
F
125a

FIG. 7A

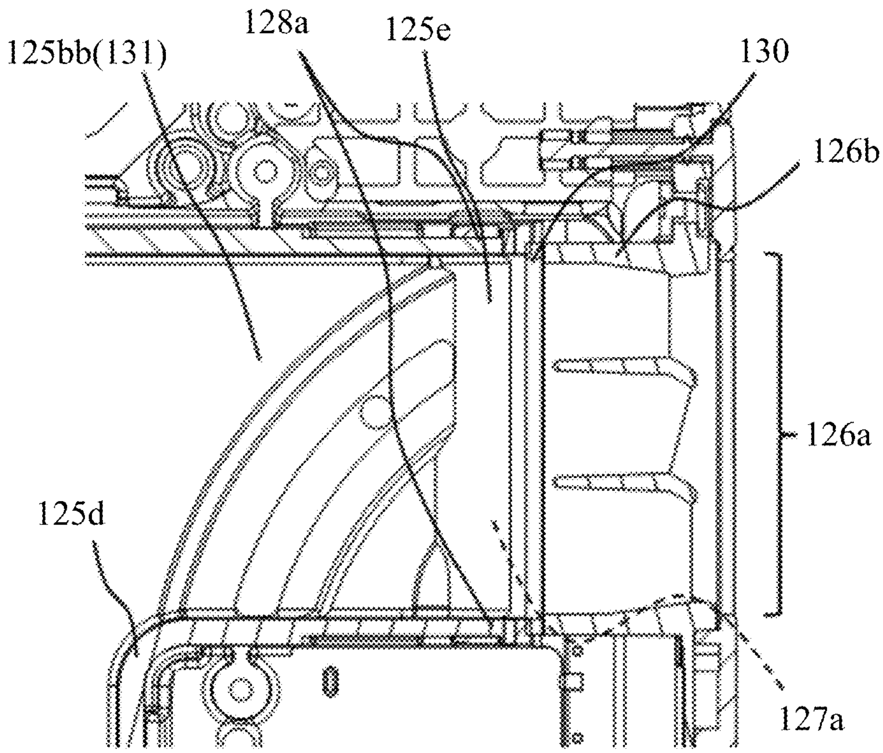

125bb(131)
128a
125e
130
126b
126a
125d
127a

FIG. 7B

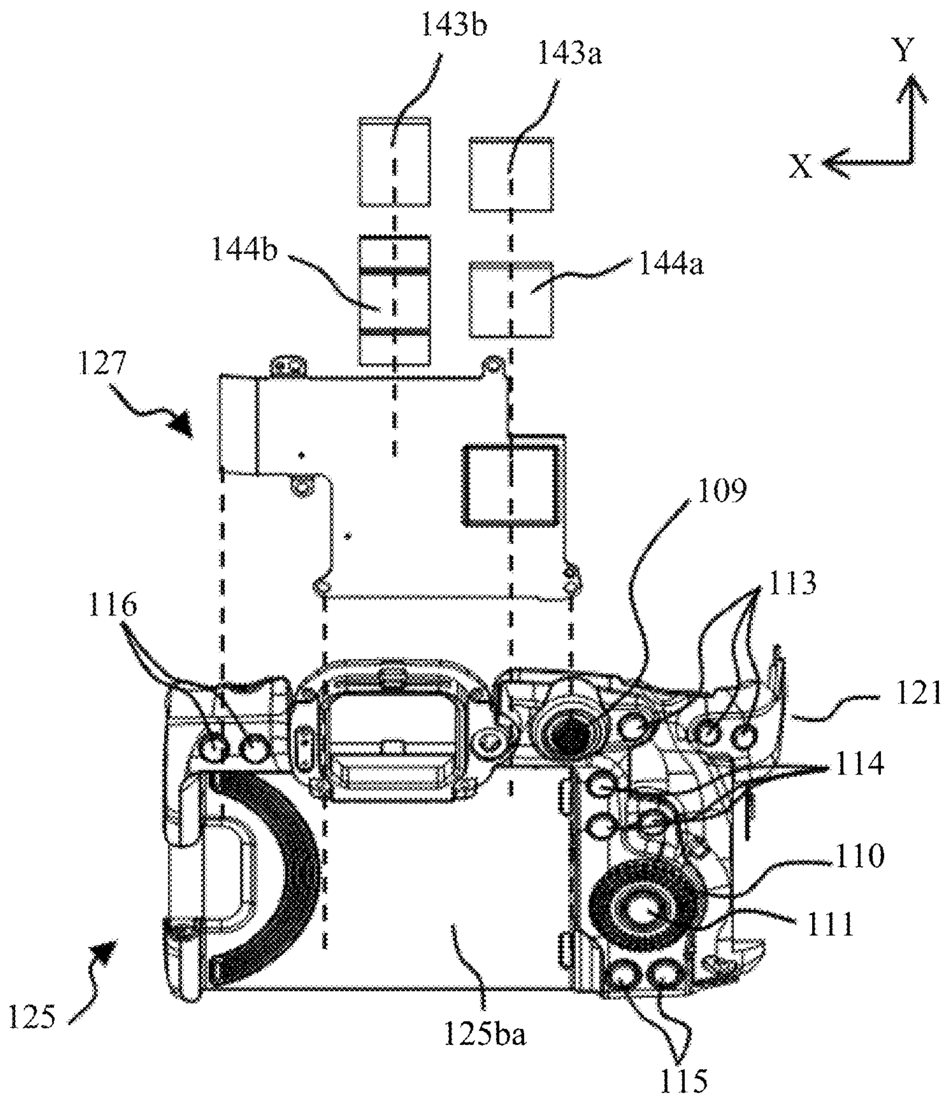

143b
143a
Y
X
144b
144a
127
109
116
113
121
114
110
111
125
125ba
115

FIG. 7C

# IMAGE PICKUP APPARATUS AND CAMERA SYSTEM HAVING THE SAME

## BACKGROUND

### Technical Field

**[0001]** The present disclosure relates an image pickup apparatus to which a cooling apparatus is detachable.

### Description of Related Art

**[0002]** In image pickup apparatuses, the calorific value has increased due to factors such as enhanced functionality leading to greater power consumption and miniaturization of electronic components, and a heat radiating structure has been provided to suppress temperature rise.

**[0003]** Japanese Patent Laid-Open No. 2019-219458 discloses a configuration for cooling the interior and exterior of an image pickup apparatus by forming an inlet port and an exhaust port on the rear cover of the image pickup apparatus, and providing a duct between the rear cover's exterior side and the interior side of a display unit arranged further outward.

**[0004]** Japanese Patent Laid-Open No. 2012-198447 discloses a configuration that includes inlet and exhaust ports leading to the interior of an image pickup apparatus and a fan unit, which, when mounted, creates an airflow within the internal space to enhance portability while cooling the interior and exterior as needed.

**[0005]** However, the configuration of Japanese Patent Laid-Open No. 2019-219458 does not allow a fan unit for directing air through a duct to be detachably mounted, making it impossible to improve cooling performance. Additionally, the arrangement of the inlet port and exhaust port is limited to within the rear cover, resulting in low flexibility in their placement.

**[0006]** In the configuration of Japanese Patent Laid-Open No. 2012-198447, the inlet port of the image pickup apparatus is provided on the tripod mounting means, limiting the placement of the inlet port. Additionally, there is no duct provided inside the image pickup apparatus, causing the air from the fan unit to easily diffuse.

## SUMMARY

**[0007]** An image pickup apparatus according to one aspect of the present disclosure is configured with a detachable cooling apparatus including a fan unit. The image pickup apparatus includes a heat generating member, a first exterior member that forms a first surface of a housing and has a first opening, a second exterior member that forms a second surface different from the first surface of the housing and has a second opening, a heat radiating member that is arranged inside the housing to face a surface opposite to the first surface and is configured to cool heat generated by the heat generating member, a first sealant that is arranged between the first exterior member and the heat radiating member, and a second sealant that is arranged between the first exterior member and the second exterior member, and between the heat radiating member and the second exterior member. The first exterior member, the second exterior member, and the heat radiating member form a first flow path of air between the first opening and the second opening. The first sealant is placed in a pressed state between the first exterior member and the heat radiating member. The second sealant is placed in a pressed state between the first exterior member, the second exterior member, and the heat radiating member.

**[0008]** Further features of various embodiments of the disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0009]** FIGS. **1**A and **1**B are external views of an image pickup apparatus according to the present embodiment.

**[0010]** FIGS. **2**A and **2**B are explanatory diagrams of an external cooling accessory.

**[0011]** FIGS. **3**A and **3**B are external views of a camera body and the external cooling accessory.

**[0012]** FIGS. **4**A and **4**B are system diagrams of the camera body and the external cooling accessory.

**[0013]** FIGS. **5**A to **5**D are explanatory diagrams of the interior of the camera body.

**[0014]** FIGS. **6**A and **6**B are side views of the camera body.

**[0015]** FIGS. **7**A to **7**C are internal views of the camera body.

## DETAILED DESCRIPTION

**[0016]** Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure. Corresponding elements in respective figures will be designated by the same reference numerals, and a duplicate description thereof will be omitted.

**[0017]** FIGS. **1**A and **1**B are external views of a camera body **100**, which serves as an image pickup apparatus according to the present embodiment. FIG. **1**A illustrates the camera body **100** viewed from the front and FIG. **1**B illustrates the camera body **100** viewed from the rear. In the present embodiment, the width direction of the camera body **100** is defined as the X direction, the height direction as the Y direction, and the direction perpendicular to the image pickup surface of an image pickup element (described later), corresponding to the optical axis direction of an unillustrated imaging optical system, as the Z direction.

**[0018]** A rear display unit **101** is attached to the rear of the camera body **100**, allowing it to be opened, closed, and rotated relative to the camera body **100**, and displays images generated by imaging, as well as various types of information related to imaging. The rear display unit **101** is equipped with a touch panel capable of detecting touch operations by a user (photographer) on its display surface (operating surface). An upper display unit **102** is provided on the upper surface of the camera body **100** and is capable of displaying the settings of various imaging parameters, such as shutter speed and aperture.

**[0019]** A shutter button **103** is an operation member that the user operates when instructing the camera body **100** to capture an image. A mode selection switch **104** is an operation member that the user operates when switching between various modes. A terminal cover **105** is a cover that protects a connector capable of connecting to a cable extending from an external device. A main electronic dial **106** is an operation member that the user rotates when changing the setting values of imaging parameters. A power switch **107** is an operation member that the user operates to toggle the power of the camera body **100** ON or OFF. A sub-electronic dial **108** is an operation member that the user operates to move selection frames, such as autofocus (AF) frames, or to scroll through images.

**[0020]** A multi-controller **109** is provided on the rear of the camera body **100** and is configured to allow input through both pressing the key top and tilting it in upward, downward, leftward, rightward, or diagonal directions. By operating the multi-controller **109**, the user can move the selection frame or select items from various menus.

**[0021]** A rear electronic dial **110** is an operation member that the user operates to move the selection frame or scroll through images. The rear electronic dial **110** is arranged to allow the user to operate it intuitively while replaying captured images on the rear display unit **101**, and it is also conveniently located for operation when the user holds the camera body **100** in a vertical orientation. A SET button **111**, located at the center of the rear electronic dial **110**, is a push button that serves as an operation member for the user to confirm selected items or perform similar actions.

**[0022]** A video button **112** is an operation member that the user operates to start and stop video recording. A first button group **113** is an operation member related to focus and exposure, and includes the AF start button, AE lock button, and AF frame selection button. By pressing a button included in the first button group **113** while in the imaging standby state, the user can start AF, change the AF frame, or lock the exposure.

**[0023]** A second button group **114** includes the zoom in/out button, the information display button, and the quick settings button. By operating the zoom in/out button in live view mode during imaging, the user can toggle the zoom of the live view image on and off. Additionally, in playback mode, by operating the zoom in/out button, the user can toggle the zoom of the playback image on and off. By operating the information display button, the user can switch the display method of the information shown on the rear display unit **101**. By operating the quick settings button, the user can transition the display on the rear display unit **101** to the screen for changing imaging parameters.

**[0024]** A third button group **115** includes the playback button and the delete button. By operating the playback button, the user can switch between imaging mode and playback mode. When the playback button is operated in imaging mode, the camera transitions to playback mode, displaying the most recent captured image stored on the media (not illustrated) on the rear display unit **101**. In playback mode, when the user selects a captured image and operates the delete button, the selected image can be deleted.

**[0025]** A fourth button group **116** includes the menu button and the rating button. By operating the menu button, the user can display a menu screen on the rear display unit **101** showing configurable items. The user can intuitively select items and change settings by touching the menu screen displayed on the rear display unit **101** or by operating the multi-controller **109**, rear electronic dial **110**, and SET button **111**. In playback mode, by operating the rating button, the user can assign a rating to the playback image.

**[0026]** A mount section **117** is configured to allow the mounting and removal of an interchangeable lens (not illustrated) equipped with an imaging optical system. A communication terminal **118** is provided inside the mount section **117** and is used for communication between the camera body **100** and the interchangeable lens.

**[0027]** A viewfinder **119** is an electronic viewfinder provided at the upper rear of the camera body **100**. By looking through the viewfinder **119**, the user can view the live view image and other displayed images. An eyepiece cover **120** is a rubber component that contacts the user's face around the eyes when looking through the viewfinder **119**.

**[0028]** A grip section **121** is a handle component with a shape that allows the user to easily grip the camera body **100** with their right hand.

**[0029]** A card cover **122** is a cover for a media slot **153**, which stores the media (not illustrated), and is located where the user's palm would contact the grip section **121**.

**[0030]** A tripod socket **123** is an attachment component provided on the bottom of the camera body **100**, used for mounting external accessories, such as an external cooling accessory **200**, which can be attached and detached from the camera body **100**, as described later.

**[0031]** A battery cover **124** is a cover for storing the battery (not illustrated) and is detachably mounted to a bottom exterior member **132**.

**[0032]** A rear exterior member (first exterior member) **125** is an exterior member that forms the rear (first surface) of the camera body **100**'s housing and is molded from materials such as metal or polycarbonate. The rear exterior member **125** includes a main body inlet port (first opening) **125*a*** provided on the bottom side of the camera body **100** and a storage appearance surface (storage surface) **125*ba***, which is part of the rear of the camera body **100** and houses the rear display unit **101**.

**[0033]** A side exterior member (second exterior member) **126** is an exterior member that forms the left side (second surface) of the camera body **100**'s housing when viewed from the rear of the camera body **100** and is molded from materials such as metal or polycarbonate. The side exterior member **126** includes a main body exhaust port **126*a***.

**[0034]** The bottom exterior member **132** is an exterior member that forms the bottom of the camera body **100**'s housing and is molded from materials such as metal or polycarbonate.

**[0035]** FIGS. **2**A and **2**B are explanatory diagrams of the external cooling accessory (cooling apparatus) **200**, which is detachable from the camera body **100**. FIG. **2**A is a front perspective view of the external cooling accessory **200**, and FIG. **2**B is a cross-sectional view along line A-A of FIG. **2**A.

**[0036]** The external cooling accessory **200** includes an accessory exterior **201**, tripod screw **202**, tripod screw operation section **203**, fan (fan unit) **204**, accessory inlet port **205**, accessory exhaust port **206**, and accessory cushion member (third sealant) **207**. By operating the tripod screw operation section **203** and rotating the tripod screw **202**, the camera body **100** and the external cooling accessory **200** can be secured together. The accessory cushion member **207** is provided on the upper surface of the accessory exhaust port **206**.

**[0037]** By driving the fan **204**, a second duct (second flow path) **208** is formed inside the external cooling accessory **200**, extending from the accessory inlet port **205**, illustrated by arrow F, through the fan **204** to the accessory exhaust port **206**. A removable battery (not illustrated) is installed inside the external cooling accessory **200**. The external cooling accessory **200** is equipped with a power terminal **252*a*** and a communication terminal **252*b***. The power terminal **252*a*** enables power supply between the camera body **100** and the external cooling accessory **200**. The communication terminal **252***b* allows communication between the camera body **100** and the external cooling accessory **200**.

**[0038]** FIGS. **3**A and **3**B are external views of the camera body **100** and the external cooling accessory **200**. FIG. **3**A illustrates the camera body **100** viewed from the front when the external cooling accessory **200** is not attached. FIG. **3**B illustrates the camera body **100** as part of a camera system, including the external cooling accessory **200** when attached, viewed from the front.

**[0039]** The state illustrated in FIG. **3**A is used when cooling inside the camera body **100** is not required. Since the external cooling accessory **200** is not required, it offers the advantage of improved portability for users who do not need internal cooling of the camera body **100**.

**[0040]** The external cooling accessory **200** is secured by removing the battery cover **124** from the camera body **100** and engaging the tripod socket **123** with the tripod screw **202**. When the fan **204** is operated, a flow path is formed from the accessory inlet port **205** to the main body exhaust port **126***a*, as illustrated by arrow F. This allows for cooling of the heat inside the camera body **100** while the fan **204** is operating.

**[0041]** As described above, the external cooling accessory **200** should only be attached to the camera body **100** when internal cooling is necessary. By providing the option to attach or detach the external cooling accessory **200** based on the user's needs, it is possible to balance the portability of the camera body **100** with the cooling performance when required.

**[0042]** FIGS. **4**A and **4**B are system diagrams of the camera body **100** and the external cooling accessory **200**. FIG. **4**A is a system diagram of the camera body **100** in the state illustrated in FIG. **3**A, and FIG. **4**B is a system diagram of the camera body **100** and the external cooling accessory **200** in the state illustrated in FIG. **3**B.

**[0043]** A main circuit board **142** is a printed wiring board (PWB) on which various components, including a CPU **143**, a heat generating member, are mounted. The CPU **143** performs overall operation control of the camera body **100**, executing control of each functional block and performing necessary calculations based on the computer program loaded from a memory **152**. A power supply **150** provides electrical power to each circuit within the camera body **100**. An image pickup element **141**, which is made up of a CCD or CMOS sensor, converts the optical image of the subject into a photoelectric signal and outputs an image signal. The image signal output from the image pickup element **141** is converted into image data by an image processing unit **151** and output to the CPU **143**. A shutter **156** is arranged in front of the image pickup element **141** and adjusts the exposure time of the image pickup element **141**. A shutter control unit **154** drives the shutter **156** based on signals input from the CPU **143**. An operation detection unit **157** outputs a signal to the CPU **143** when the mode selection switch **104** is operated by the user, changing the shooting conditions such as exposure and shutter speed.

**[0044]** The user can select a desired video recording mode from a plurality of multiple video recording modes by operating the mode selection switch **104**. Video recording modes include, for example, high-quality mode and low-quality mode. In low-quality mode, the calorific value of electronic components such as the image pickup element **141** and CPU **143** is small, allowing for longer video recording times. On the other hand, in high-quality mode, the processing load on components like the image pickup element **141** and CPU **143** is large, resulting in a higher calorific value for the electronic components and the media (not illustrated). When the calorific value becomes large and exceeds the allowable temperature, it can cause thermal runaway or a reduction in product lifespan, forcing the video recording time to be shortened. In such cases, the external cooling accessory **200** can be attached to the camera body **100**.

**[0045]** The external cooling accessory **200** is equipped with a power supply **250**, a control substrate **251**, and a connection terminal **252**, and is connected to the camera body **100** for power supply and communication via the connection terminal **252** and a connection terminal **155** provided on the camera body **100**. The power supply **250** supplies power to each circuit in the camera body **100** through the connection terminal **252**. The control substrate **251** drives the fan **204** based on signals input from the CPU **143**.

**[0046]** The internal configuration and cooling method of the camera body **100** are explained below. FIGS. **5**A to **5**D are explanatory diagrams of the interior of the camera body **100**. FIG. **5**A is an exploded perspective view of the interior of the camera body **100** viewed from the front when the external cooling accessory **200** is not attached. FIG. **5**B is a perspective view of the interior of the camera body **100** viewed from the front when the external cooling accessory **200** is attached. FIG. **5**C is a cross-sectional view along line B-B of FIG. **5**B. FIG. **5**D is an enlarged partial view of section C in FIG. **5**C.

**[0047]** The rear exterior member **125** includes a storage inner surface **125***bb*, which is the backside of the storage appearance surface **125***ba*, and a wall section **125***c* formed to protrude in the Z-direction from the storage inner surface **125***bb*. A metal sheet member (heat radiating member) **127** is arranged inside the housing, facing the rear side of the camera body **100** (surface on the storage appearance surface **125***ba* side of the rear exterior member **125**) and the opposite side (surface on the storage inner surface **125***bb* side of the rear exterior member **125**). The metal sheet member **127** has an outer perimeter formed along the wall section **125***c* of the rear exterior member **125**. In this embodiment, the metal sheet member **127** is assumed to be made of aluminum alloy, but this is not a limitation. A first cushion member (first sealant) **128** is arranged between the metal sheet member **127** and the wall section **125***c* and is applied to the wall section **125***c*. The metal sheet member **127** is fixed to the wall section **125***c* in the Z-direction through the first cushion member **128** by a fastening member **129**.

**[0048]** A first CPU (second heat generating member) **143***a* and a second CPU (first heat generating member) **143***b* are arranged on the metal sheet member **127** side of the main circuit board **142** in the Z-direction. The first CPU **143***a* and the second CPU **143***b* have different calorific values. In this embodiment, the first CPU **143***a* has a smaller calorific value than the second CPU **143***b*. A first heat radiating member (second heat radiating member) **144***a* is arranged between the metal sheet member **127** and the first CPU **143***a*, and a second heat radiating member (first heat radiating member) **144***b* is arranged between the metal sheet member **127** and the second CPU **143***b*. The first heat radiating member **144***a* is thermally in contact with the first CPU **143***a* and transfers heat emitted from the first CPU **143***a* to the metal sheet member **127**. The second heat radiating member **144***b* is thermally in contact with the second CPU **143***b* and transfers heat emitted from the second CPU **143***b* to the metal sheet member **127**. In this embodiment, the first heat radiating member **144***a* is assumed to be made of thermal conductive rubber, and the second heat radiating member **144***b* is assumed to be a graphite sheet, but this is not a limitation. The side exterior member **126** includes a protruding section **126***b* that projects in the X-direction from the rear surface of the surface where the main body exhaust port **126***a* is provided. A second cushion member (second sealant) **130** is applied to the tip surface of the protruding section **126***b*.

**[0049]** In this embodiment, by dividing the main body inlet port **125***a* into the rear exterior member **125** and the main body exhaust port **126***a* into the side exterior member **126**, the size increase of the rear exterior member **125** can be suppressed. As a result, it is possible to reduce the weight of the camera body **100**. Additionally, by forming the main body inlet port **125***a* and the main body exhaust port **126***a* in separate exterior members, the freedom in positioning the main body inlet port **125***a* and main body exhaust port **126***a* is increased.

**[0050]** In this embodiment, the main body inlet port **125***a* is provided on the bottom surface, which has fewer operation members, as described in FIGS. **1**A and **1**B, compared to the top and rear surfaces, and the main body exhaust port **126***a* is provided on the left side surface when viewed from the rear of the camera body **100**. This allows the user to operate the camera body **100** without blocking the main body inlet port **125***a* or the main body exhaust port **126***a*.

**[0051]** The thickness of the first cushion member **128** is greater than the clearance in the Z-direction between the wall section **125***c* and the metal sheet member **127**. As a result, when the metal sheet member **127** is secured to the wall section **125***c* by the fastening member **129**, the first cushion member **128** is always pressed in the Z-direction. With this configuration, a first duct (first flow path) **131** is formed in the camera body **100**, extending from the main body inlet port **125***a* to a merging section **127***a* via the outer peripheral shape of the metal sheet member **127**. The first duct **131** ensures internal sealing by pressing the first cushion member **128** and the second cushion member **130**.

**[0052]** The accessory cushion member **207** is arranged to overlap with the outer periphery of the main body inlet port **125***a* in the Y-direction. The thickness of the accessory cushion member **207** is greater than the clearance in the Y-direction between the accessory exterior **201** and the rear exterior member **125** as well as the bottom exterior member **132**. As a result, when the external cooling accessory **200** is attached to the camera body **100**, the accessory cushion member **207** is always pressed in the Y-direction. With this configuration, it becomes possible to ensure sealing from the accessory exhaust port **206** to the main body inlet port **125***a* during the operation of the fan **204** and to form the flow path illustrated by arrow F in the figure, extending from the accessory inlet port **205** to the first duct **131**. To sufficiently secure this flow path, the clearance in the Z-direction between the storage inner surface **125***bb* and the metal sheet member **127**, which constitutes the first duct **131**, is made greater than the thickness of the metal sheet member **127**.

**[0053]** The heat generated by the first CPU **143***a* and the second CPU **143***b* is transferred to the metal sheet member **127** via the first heat radiating member **144***a* and the second heat radiating member **144***b*, respectively. The heat of the metal sheet member **127** is cooled by the air flowing from the accessory inlet port **205** to the first duct **131**. In shooting modes where the heat sources of the camera body **100**, including the first CPU **143***a* and the second CPU **143***b*, have low calorific value, the external cooling accessory **200** is not attached. On the other hand, in shooting modes with high calorific value, a camera system is configured by attaching the external cooling accessory **200** and driving the fan **204**. This configuration ensures the portability of the camera body **100** while improving heat dissipation and providing users with greater flexibility in operation.

**[0054]** FIGS. **6**A and **6**B are side views of the camera body **100**. FIG. **6**A illustrates the state with the terminal cover **105** removed. FIG. **6**B illustrates the state with the terminal cover **105** attached.

**[0055]** The camera body **100** includes a USB terminal (first connector) **145** and an HDMI® terminal (second connector) **146**. The USB terminal **145** and the HDMI terminal **146** are arranged to be exposed through an opening in the side exterior member **126**. The USB terminal **145** and the HDMI terminal **146** are protected by the terminal cover **105** when not in use. In this embodiment, the HDMI terminal **146** has a larger area (projected area) on the side exterior member **126** compared to the USB terminal **145**. The main body exhaust port **126***a* is arranged rearward in the Z direction of the USB terminal **145** and upward in the Y direction of the HDMI terminal **146**. By utilizing the space created by the difference in area between the USB terminal **145** and the HDMI terminal **146** to position the main body exhaust port **126***a*, it is possible to prevent the terminal cover **105** from expanding rearward in the Z direction and to reduce the size and weight of the camera body **100** in the Z. The USB terminal **145** and the HDMI terminal **146** may be mounted on the main circuit board **142** or on an unillustrated circuit board arranged parallel to the main circuit board **142**.

**[0056]** FIGS. **7**A to **7**C are internal views of the camera body **100**. FIG. **7**A is a cross-sectional view along line D-D of FIG. **6**B. FIG. **7**B is an enlarged view of section E in FIG. **7**A. FIG. **7**C is an exploded perspective view of the interior of the camera body **100** as viewed from the rear.

**[0057]** In this embodiment, the calorific value of the second CPU **143***b* is greater than that of the first CPU **143***a*. The wall section **125***c* forms the wall of the first duct **131** and includes a bent part **125***d* that forcibly redirects air entering from the main body inlet port **125***a* toward the main body exhaust port **126***a*. The second CPU **143***b* is arranged near the main body inlet port **125***a* side relative to the bent part **125***d*. Generally, the pressure is higher on the outer side of a flow path and lower on the inner side, resulting in higher flow velocity on the inner side and lower flow velocity on the outer side. Inside the first duct **131**, as illustrated by arrow F, a flow path is formed where the side near the second CPU **143***b* corresponds to the higher-velocity inner side, and the side near the first CPU **143***a* corresponds to the lower-velocity outer side. As described above, by increasing the flow velocity near the second CPU **143***b*, which has a higher calorific value, it is possible to efficiently enhance the heat dissipation of the second CPU **143***b* within the limited flow path.

**[0058]** The second cushion member **130** is arranged in the X direction between a merging section **125***e* of the rear exterior member **125** and the protruding section **126***b* of the side exterior member **126**, between the merging section **127***a* of the metal sheet member **127** and the protruding section **126***b*, and between a merging section **128***a* of the first cushion member **128** and the protruding section **126***b*.

**[0059]** When the side exterior member **126** is attached to the camera body **100** in the X direction, the thickness of the second cushion member **130** is larger than the X-direction clearance between the protruding section **126***b* and the merging sections **125***e*, **127***a*, and **128***a*. As a result, when the side exterior member **126** is attached to the camera body **100** in the X direction, the second cushion member **130** is always pressed in the X direction. With this configuration, the sealing between the first duct **131** and the main body exhaust port **126***a* during the operation of the fan **204** is ensured, and as illustrated in FIG. **7**A, a flow path illustrated by arrow F from the main body inlet port **125***a* to the main body exhaust port **126***a* is formed.

**[0060]** The multi-controller **109**, rear electronic dial **110**, SET button **111**, buttons **113** to **116** (first to fourth), and grip section **121** are arranged in positions that do not overlap with the first duct **131** when viewed from the Z direction. Furthermore, the first CPU **143***a*, which has a smaller calorific value compared to the second CPU **143***b*, is arranged closer to the grip section **121**. In this embodiment, the thermal conductivity of the first heat radiating member **144***a* is set to be smaller than that of the second heat radiating member **144***b*. In this way, the first CPU **143***a* and second CPU **143***b*, which are heat sources, as well as the metal sheet member **127** that receives their heat, are intentionally arranged away from the components that the user may touch, both in the X and Y directions.

**[0061]** With the above configuration, the heat dissipation of the first CPU **143***a* and the second CPU **143***b* through the airflow from the fan **204** can be performed efficiently, while preventing the rear exterior member **125** from becoming excessively hot in a short period.

**[0062]** While the disclosure has described example embodiments, it is to be understood that the disclosure is not limited to the example embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

**[0063]** According to the present disclosure, it is possible to provide an image pickup apparatus that achieves both portability and cooling performance while enhancing the flexibility of inlet port and exhaust port placement.

**[0064]** This application claims priority to Japanese Patent Application No. 2024-031873, which was filed on Mar. 4, 2024, and which is hereby incorporated by reference herein in its entirety.

What is claimed is:

**1**. An image pickup apparatus to which a cooling apparatus including a fan unit is detachable, the image pickup apparatus comprising:

a heat generating member;

a first exterior member that forms a first surface of a housing and has a first opening;

a second exterior member that forms a second surface different from the first surface of the housing and has a second opening;

a heat radiating member that is arranged inside the housing to face a surface opposite to the first surface and is configured to cool heat generated by the heat generating member;

a first sealant that is arranged between the first exterior member and the heat radiating member; and

a second sealant that is arranged between the first exterior member and the second exterior member, and between the heat radiating member and the second exterior member,

wherein the first exterior member, the second exterior member, and the heat radiating member form a first flow path of air between the first opening and the second opening,

wherein the first sealant is arranged in a pressed state between the first exterior member and the heat radiating member, and

wherein the second sealant is arranged in a pressed state between the first exterior member, the second exterior member, and the heat radiating member.

**2**. The image pickup apparatus according to claim **1**,

wherein the first surface is a rear surface of the image pickup apparatus, and

wherein the second surface is a side surface of the image pickup apparatus.

**3**. The image pickup apparatus according to claim **1**, further comprising:

an image pickup element configured to photoelectrically convert an optical image of a subject; and

a first connector and a second connector that are provided on the second surface and are configured to connect an external device,

wherein the second connector is formed with a larger projected area than the first connector, and

wherein the second opening is arranged to overlap the second connector in a direction perpendicular to an image pickup surface of the image pickup element.

**4**. The image pickup apparatus according to claim **1**, wherein the cooling apparatus is detachable from a bottom surface of the image pickup apparatus.

**5**. The image pickup apparatus according to claim **1**, further comprising:

an image pickup element configured to photoelectrically convert an optical image of a subject;

an operation member operated by a photographer; and

a gripping member gripped by the photographer,

wherein the operation member and the gripping member are arranged not to overlap the first flow path when viewed in a direction perpendicular to an image pickup surface of the image pickup element.

**6**. The image pickup apparatus according to claim **1**, further comprising:

a first heat generating member;

a second heat generating member with a smaller calorific value than the first heat generating member;

a first heat radiating member thermally in contact with the first heat generating member; and

a second heat radiating member thermally in contact with the second heat generating member;

wherein the first flow path is thermally in contact with the first heat radiating member and the second heat radiating member,

wherein the first flow path includes a bent part, and

wherein the first heat generating member is arranged closer to the bent part than the second heat generating member.

**7**. The image pickup apparatus according to claim **6**, further comprising a gripping member gripped by a photographer, wherein the first heat generating member is arranged farther from the gripping member than the second heat generating member.

**8**. The image pickup apparatus according to claim **1**, further comprising a display configured to display various types of information,

wherein a part of the first surface has a storage surface formed to house the display, and

wherein the first flow path is formed by a surface opposite to the storage surface, the second exterior member, and the heat radiating member.

**9**. A camera system comprising:

an image pickup apparatus;

a cooling apparatus that includes a fan unit and is detachable from the image pickup apparatus,

wherein the image pickup apparatus includes:

a heat generating member;

a first exterior member that forms a first surface of a housing and has a first opening;

a second exterior member that forms a second surface different from the first surface of the housing and has a second opening;

a heat radiating member that is arranged inside the housing to face a surface opposite to the first surface and is configured to cool heat generated by the heat generating member;

a first sealant that is arranged between the first exterior member and the heat radiating member; and

a second sealant that is arranged between the first exterior member and the second exterior member, and between the heat radiating member and the second exterior member,

wherein the first exterior member, the second exterior member, and the heat radiating member form a first flow path of air between the first opening and the second opening,

wherein the first sealant is placed in a pressed state between the first exterior member and the heat radiating member, and

wherein the second sealant is placed in a pressed state between the first exterior member, the second exterior member, and the heat radiating member.

**10**. The camera system according to claim **9**, wherein the cooling apparatus includes a third opening formed on a third surface that faces a surface where the first opening is formed, a fourth opening formed on a fourth surface different from the third surface of the cooling apparatus, and a third sealant arranged around an outer periphery of the third opening, in a pressed state between the surface where the first opening is formed and the third surface.

**11**. The camera system according to claim **10**,

wherein, inside the cooling apparatus, a second flow path of air is formed from the fourth opening to the third opening via the fan unit, and

wherein the first flow path and the second flow path are connected via the first opening and the third opening.

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